Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Sb2O3-free flame-retardant synergists for polymers*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		*Chemicals*	
	Medium Category	Small Category	Medium Category	Small Category
	Polymeric Materials	*Special Functional Materials Technology*	*Fine Chemicals*	*Additives*
Concept of Proposed Technology	*Replace antimony trioxide (Sb2O3) as a flame-retardant synergist with antimony-free packages that combine nitroxyl-generating hindered amine light stabilizers (N-alkoxy HALS), phosphorus flame retardants (P-FRs: phosphinates/-ites/-ones), and halogen-free sulfonic/sulfinic esters; optionally low-migration oxyimide co-additives for condensed-phase char support. The designed packages act in both gas and condensed phases, raising UL 94 performance, lowering peak heat release and smoke, while maintaining processing and mechanical properties.* *Main functions:* • *Flame inhibition in gas phase (radical quenching via nitroxyl/HBr-free pathways and P-species).* • *Condensed-phase char promotion (intuminescence) and destabilization of pyrolysis toward less flammable volatiles; reduced smoke and corrosivity.* • *Process stability and recyclability: low volatility/migration; limited MVR drift across reprocessing*			

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	cycles. *Potential uses: Electrical/electronic housings and components; cable ducts/connectors; automotive interior/exterior trim and under-hood plastics; white goods; consumer electronics; selected building/plenum parts (subject to local codes).*
Rationale for Joint R&D Project	*Technical need: Industry seeks antimony-free and/or completely halogen-free systems due to EHS/ESG and regulatory pressure (RoHS/WEEE), without sacrificing UL 94 ratings or increasing total FR loading and smoke. Achieving balanced gas/condensed-phase action at lower dosages in diverse polymer families requires mechanism-driven design and pilot-scale validation.* *Collaboration rationale:* • *Fraunhofer LBF: Expertise in additive design, mechanistic kinetics (TGA/DSC/ESR/FTIR; cone calorimetry), formulation screening, durability/LCA; IP on oxyimides, sulfon/sulfinic esters, and stabilization systems.* • *Korean SMEs: Fine-chemical synthesis/scale-up, compounding, pilot line processing, and conformity testing; rapid path to local certification and market access.* • *Joint R&D integrates design-to-scale-up, aligns to Korean standards pathways (UL/IEC/ISO with local test orgs), and accelerates commercialization in Advanced Mobility supply chains.*
Key Words	1. *Antimonytrioxide/ATO/Sb-free synergists* 2. *Nitroxyl-generating HALS* 3. *Phosphorus flame retardants (P-FRs)* 4. *Sulfon/sulfinic ester radical generators* 5. *Low-migration oxyimides*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4-5)	End of the Project	Level (6-7)
R&D Objectives	• *Tunable HALS derivatives (N-alkoxy/NOR-type) optimized for 180–240°C decomposition windows, low volatility/migration in PP/PA/PBT/ABS.* • *Halogen-free sulfonic/sulfinic esters as dual-use FR synergists and melt radical generators with controlled T½/Ton and polymer compatibility.* • *Oxyimide and phosphorus-functional oxyimide co-additives (mono-/oligo-/polymeric) for condensed-phase char and potential dual stabilization.* • *Synergistic P-FR packages (Al/Zn phosphinates/-ites/-ones) with the above co-additives for reduced total loading and improved smoke/toxicity profile.* • *Masterbatch routes and processing window maps for injection molding/extrusion; reprocessing stability protocols for recyclability.*			
R&D Target Specifications	*UL 94:* • *PA6/66, PBT: V-0 at 0.8 to 1.6 mm; self-extinguish with no afterflame drip ignition.* • *PP (filled/unfilled, as applicable): V-2 at 1.6–3.2 mm; V-0 target for selected PP formulations at 3.2 mm.* *Limiting Oxygen Index (ISO 4589-2): +3 vol% vs polymer baseline (or ≥28% for PA/PBT targets).* *Cone calorimetry (ISO 5660):* • *Peak Heat Release Rate (PHRR): ≥30% reduction versus Sb2O3-based reference compound.* • *Smoke Specific Extinction Area (SEA): ≥20% reduction versus reference.* *Migration/VOC/fog (interior parts): meets automotive OEM VOC/fog limits; Δcolor ≤1.0 ΔE* after heat aging 1000 h at application-relevant conditions.* *Processing/recycling stability:* • *ΔMVR ≤10% after 3× re-extrusion cycles at defined shear/thermal history.* • *Mechanical retention (E-modulus/impact): ≥90% of baseline (non-FR) for selected formulations at target loadings.*			
R&D Plan	*Approach* • *Mechanism-led formulation design drawing on Marney et al. (2007/2008) insights (gas/condensed-phase action of FRs with N-alkoxy HALS), extended to antimony-free packages with P-FR synergy and halogen-free sulfon/sulfinic esters; optional oxyimide co-FR/stabilizers. Iterative screening and scale-up in collaboration with Korean SMEs.* *Key tasks and methodologies* • *Additive design & analytics: TGA/DSC for Ton/T½; ESR for radical generation; thin-film FTIR for bond evolution; TDe-GC-MS for volatile mapping; mechanistic modeling (Horowitz–Metzger kinetics).* • *Formulation screening in PP, PA6/66, PBT, PC/ABS, HIPS/ABS, TPU; compounding masterbatch routes; dispersion and rheology control.* • *Flammability & fire testing: UL 94 (IEC 60695-11-10), LOI/NOI; cone calorimetry (ISO 5660); dripping behavior, smoke/CO–CO2 analysis.* • *Processing & durability: injection/extrusion windows, reprocessing cycles, migration/VOC/fog, color stability; mechanical/thermal aged properties.* • *Scale-up & certification: pilot production, conformity tests via KR test organizations, application demos (E&E housings, cable channels, automotive interior parts).* • *LCA/ESG & compliance: antimony-free/halogen-free options; RoHS/WEEE alignment; cost/performance modeling; tech transfer/IP management.*			

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Timeline and milestones (3 years)

Phase	Months	Milestones	Highlights
I	0–2	M1: Kick-off & risk plan	Polymer/application prioritization; test plans; compliance criteria set
I	0–6	M2: Additive/formulation library; M3: Top-10 candidates down-selected	TGA/DSC/ESR/FTIR/TDe-GC-MS; UL 94/LOI/cone quick screens
II	7–12	M4: Pilot masterbatch; M5: Processing window maps	Scale-up with SMEs; injection/extrusion trials; initial VOC/fog checks
II	13–18	M6: Application demo parts (E&E/auto)	Full UL/IEC/ISO tests; mechanical & aging; ΔMVR after reprocessing
II	19–24	M7: Certification readiness; LCA/ESG report	Conformity test dossiers; antimony-free declarations; cost/performance
II	25–30	M8: OEM-style validation (spec alignment)	Durability, migration, color, and smoke metrics verified
II	31–36	M9: Tech transfer & commercialization plan	Final data packages; IP strategy; go-to-market with Korean SMEs

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• *Technical contributions: additive concept design; mechanistic/kinetic studies (TGA/DSC/ESR/FTIR/TDe-GC-MS); formulation screening; cone calorimetry; UL/LOI pre-screens; durability/LCA; reprocessing stability; migration/VOC/fog protocols.* • *Research scope: PP, PA6/66, PBT, PC/ABS, HIPS/ABS, TPU; dose/dispersion optimization; masterbatch strategies.* • *Resources: compounding and fire test facilities; analytics; IP portfolio on oxyimides, sulfon/sulfinic esters, stabilization systems; project management; data management for certification.* • *Specific tasks: design-of-experiments; top-candidate selection; pilot support; documentation for Phase II application; ESG compliance evidence.*
Expected R&R of Korean SME Partner	• *Technical contributions: synthesis/scale-up of HALS derivatives and sulfon/sulfinic esters; masterbatch and compounding pilot lines; tooling and part fabrication; local conformity testing.* • *Development tasks: processing window mapping; VOC/fog/migration tests per local/industry protocols; application-specific design trials.* • *Testing/validation: coordinate UL/IEC/ISO tests with KR test organizations; provide data for Phase I/II reviews; OEM-style validation in KR ecosystem.* • *Resource commitments: pilot equipment, materials procurement, local regulatory liaison; support LCA/cost modeling; participate in IP/tech-transfer and commercialization planning.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• *Technological achievements: validated antimony-free FR synergist packages hitting UL 94 V-0/V-2 targets with lower smoke; robust mechanism-backed recipes for multiple polymer families; tunable radical generators with defined decomposition windows; low-migration oxyimide co-FR options.* • *Commercialization potential: drop-in masterbatch routes; scalable supply from KR SMEs; certification-ready data packages; entry into Advanced Mobility E&E/automotive parts and export markets.* • *Broader industrial/economic impact: reduced reliance on Sb2O3 and halogens; improved EHS/ESG compliance (RoHS/WEEE); lower total FR loadings and smoke/corrosivity benefits; enhanced recyclability through stable processing and reprocessing; strengthened KR–DE SME collaboration and faster adoption in Korean supply chains.*

6. References

(1) Marney et al., Journal of Fire Sciences (2007): Mechanistic analysis of FR systems in PP (UL 94/LOI/NOI, TGA/FTIR, cone).

(2) Marney et al., Polymer Degradation & Stability (2008): Influence of N-alkoxy HALS on brominated phosphate ester FR; condensed-phase interactions (TGA/FTIR/TDe-GC-MS/ESR).

(3) WO2019020699A1: Halogen-free sulfonic/sulfinic esters as FR synergists and radical generators.

(4) WO2014154636A1 / WO2016042040A1: Oxyimide and phosphorus-functional oxyimide additives for halogen-free FR and stabilization.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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